

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Semestertoets 3
VRAESTEL

Kopiereg voorbehou

Module EBN111
Elektrisiteit en Elektronika
May 2017

Department of Electrical, Electronic and
Computer Engineering

Semester Test 3
QUESTION PAPER

Copyright reserved

Module EBN111
Electricity and Electronics
May 2017



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	46	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 min + 10 min om OKH vorm te voltooi <i>90 min + 10 min to complete OCR form</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		18 + OCR	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
All questions must be answered on the Optical Character Recognition (OCR) form.
- Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 before the start of the test.

Do step 1 in the first 90 minutes of the test

Do step 2 within the following 10 minutes.

STEP 0 (– 5 min to 0 min): Complete the front page of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2017EBN111S3**
- Use a **mechanical pencil** to complete the form so that you can erase errors.
- **Do not use commas!** Please make use of full stops.

Examples

01	–	1	.	4	5					m	A						
02	3	4	0	.	2	6				V							
03	1	5	.	2													
04	–	1	1	.	3	5	6			Ω							

INSTRUKSIES

Doen stap 0 voor die begin van die toets.

Doen stap 1 in die eerste 90 minute van die toets.

Doen stap 2 binne die daaropvolgende 10 minute.

STAP 0 (– 5 min to 0 min): Voltooi die voorblad van die OKH vorm.

STAP 1 (0 min to 90 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

STAP 2 (90 min to 100 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.

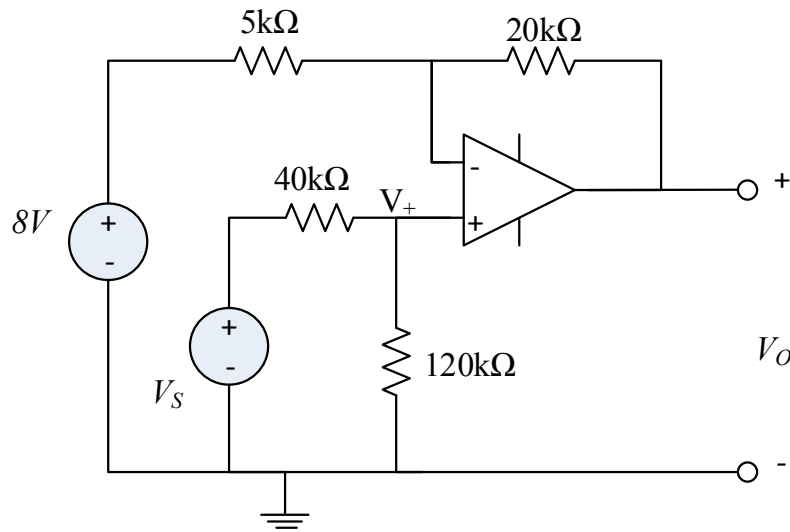
- Die “Assessment ID” is **2017EBN111S3**
- Gebruik ‘**n meganiese potlood** om antwoorde in te vul sodat jy foute kan uitvee.
- **Moenie kommas gebruik nie!** Maak eerde gebruik van punte.

Voorbeelde

01	–	1	.	4	5					m	A						
02	3	4	0	.	2	6				V							
03	1	5	.	2													
04	–	1	1	.	3	5	6			Ω							

SECTION / AFDELING 1 [4]

Question 01-02 [4]		Vraag 01-02 [4]	
Find V_O and V_S if $V_+ = 12$ V.		Bepaal V_O en V_S if $V_+ = 12$ V.	
01	V_O	01	V_O
02	V_S	02	V_S

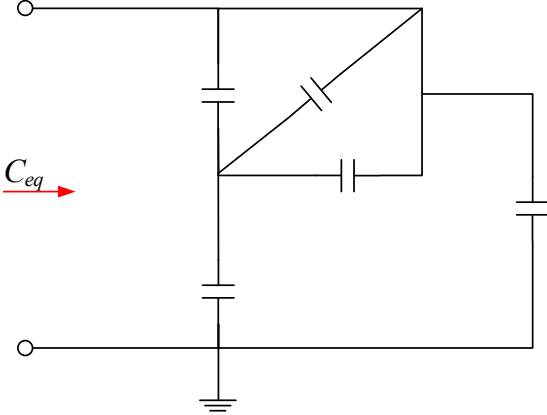


SECTION / AFDELING 2 [8]

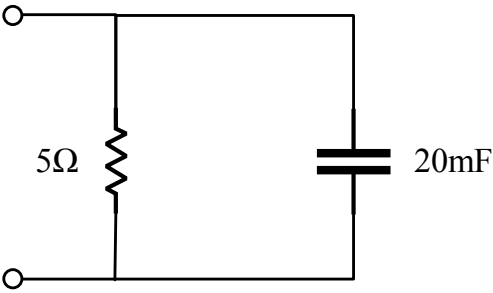
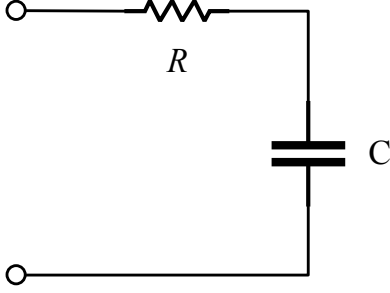
Question 03-04 [4]		Vraag 03-04 [4]	
a) Determine energy E_L stored in the inductor. b) Determine energy E_C stored in the capacitor. (The circuit is in steady-state (DC).)		a) Bepaal die energie E_L gestoor in die induktor. b) Bepaal die energie E_C gestoor in die kapasitor. (Die kringbaan is in gestadigde toestand (GS).)	
03	E_L	03	E_L
04	E_C	04	E_C

Question 05-06 [2]		Vraag 05-06 [2]	
If $v(t) = e^{-t}$ V, with $i_C(0) = 0A$ and $i_L(0) = 2A$, determine $i(t)$:		Indien $v(t) = e^{-t}$ V, met $i_C(0) = 0A$ en $i_L(0) = 2A$, bepaal $i(t)$:	
$i(t) = ae^{-t} + b$		$i(t) = ae^{-t} + b$	
05	a	05	a
06	b	06	b

The diagram shows an electrical circuit. On the left, there is a voltage source represented by a circle with a '+' sign at the top and a '-' sign at the bottom, labeled $v(t) = e^{-t}V$. This source is connected in series with a parallel combination of two components. The first component in the parallel branch is an inductor, represented by a vertical coil, labeled $0.5H$. The current through the inductor is labeled $i_L(t)$ with a downward arrow. The second component in the parallel branch is a capacitor, represented by two parallel horizontal lines, labeled $0.5F$. The current through the capacitor is labeled $i_C(t)$ with a downward arrow. The total current flowing out of the positive terminal of the voltage source is labeled $i(t)$ with a rightward arrow.

Question 07 [2]		Vraag 07 [2]	
Calculate C_{eq} if all capacitors are $10\ \mu F$.		Bepaal C_{eq} indien alle kapasitoren $10\ \mu F$ is.	
07	C_{eq}	07	C_{eq}
			

SECTION / AFDELING 3 [18]

Question 08-11 [4]		Vraag 08-11 [4]	
a) Determine R and C if the input impedances of the two circuits are exactly the same at $\omega = 50 \text{ rad/s}$.		a) Bepaal R en C vir onderstaande kringbane indien dit intree impedansie van die twee kringbane dieselfde is by $\omega = 50 \text{ rad/s}$.	
08	R	08	R
09	C	09	C
			
b) If a source of $20 \cos(2000t + 10^\circ) \text{ V}$ is connected to the series combination of a 5Ω resistor and a 2 mF capacitor, what is the current that exits the positive terminal of the source? Give the final answer in the form $I = A \angle \phi$, where $A > 0$ and $-180^\circ \leq \phi \leq 180^\circ$.		b) Indien 'n bron van $20 \cos(2000t + 10^\circ) \text{ V}$ aan die serie kombinasie van 'n 5Ω resistor en 'n 2 mF kapasitor gekoppel word, wat is die stroom wat uit die positiewe terminaal van die bron vloei? Gee die finale antwoord in die vorm $I = A \angle \phi$, met $A > 0$ en $-180^\circ \leq \phi \leq 180^\circ$.	
10	A	10	A
11	ϕ	11	ϕ

Question 12-13 [2]		Vraag 12-13 [2]	
Find the input admittance Y_T at $\omega = 100 \text{ rad/s}$. Give the final answer in rectangular form : $Y_T = Y_a + jY_b \text{ S}$ Hint: Admittance is inverse of impedance: $Y = 1/Z$ [Siemens].		Bepaal die intree admitansie Y_T by $\omega = 100 \text{ rad/s}$. Gee die finale antwoord in die reghoekige vorm : $Y_T = Y_a + jY_b \text{ S}$ Wenk: admitansie is die omgekeerde van impedansie: $Y = 1/Z$ [Siemens].	
12	Y_a	12	Y_a
13	Y_b	13	Y_b

The circuit diagram shows an input admittance Y_T looking into a network. The network starts with a 6Ω resistor in series. This is followed by a parallel branch. The first branch of the parallel section is a 1.25mF capacitor connected to ground. The second branch is a series combination of an 80mH inductor and a 6Ω resistor, also connected to ground. The input terminals are on the left, and the bottom wire is connected to a common ground symbol.

Question 14-17 [4]		Vraag 14-17 [4]	
a) Simplify the following expression of $\bar{V}$ and write the answers in rectangular form . $a + jb = \frac{(6 - j8)^2}{\sqrt[3]{27 \angle 135^\circ} + j10 \cos(60^\circ)}$ b) Determine $\bar{V}_2$ in polar form ($A \angle \theta$). $\begin{bmatrix} 3 + j4 & -j2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 3 \angle 0^\circ \\ 0 \end{bmatrix}$		a) Vereenvoudig die volgende uitdrukking en skryf die antwoord in reghoekige vorm . $a + jb = \frac{(6 - j8)^2}{\sqrt[3]{27 \angle 135^\circ} + j10 \cos(60^\circ)}$ b) Bepaal $\bar{V}_2$ in polêre vorm ($A \angle \theta$). $\begin{bmatrix} 3 + j4 & -j2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 3 \angle 0^\circ \\ 0 \end{bmatrix}$	
14	a	14	a
15	b	15	b
16	A	16	A
17	θ	17	θ

Questions 18-21 [4]		Vraag 18-21 [4]	
Given two sinusoid signals v_1 and v_2, where $v_1 = 25 \cos(15t - 30^\circ),$ $v_2 = -50 \sin(15t + 150^\circ).$ a) Determine $v = v_1 + v_2$, and write the final answer in the form $B \cos(\omega t + \phi)$ where $B > 0$ and $-180^\circ \leq \phi \leq 180^\circ$. b) Which statement is correct? 1. v_1 leads v_2 2. v_2 leads v_1 c) What is the phase angle ϕ_d ($0^\circ \leq \phi_d \leq 180^\circ$) between v_1 and v_2?		Twee sinusiedseine v_1 en v_2 word gegee: $v_1 = 25 \cos(15t - 30^\circ),$ $v_2 = -50 \sin(15t + 150^\circ).$ a) Bepaal $v = v_1 + v_2$, en skryf die finale antwoord in die vorm $B \cos(\omega t + \phi)$ met $B > 0$ en $-180^\circ \leq \phi \leq 180^\circ$. b) Watter stelling is korrek? 2. v_1 loop voor v_2 2. v_2 loop voor v_1 c) Wat is die fasehoek ϕ_d ($0^\circ \leq \phi_d \leq 180^\circ$) tussen v_1 en v_2?	
18	B	18	B
19	ϕ	19	ϕ
20	Choose 1 or 2.	20	Kies 1 of 2.
21	ϕ_d	21	ϕ_d

Question 22-25 [4]		Vraag 22-25 [4]	
a) Find $\bar{V}_{ab} = V_c + jV_d$ V if $\bar{V}_s = 100\angle 30^\circ$ V.		a) Bepaal $\bar{V}_{ab} = V_c + jV_d$ V indien $\bar{V}_s = 100\angle 30^\circ$ V.	
b) Find $\bar{I}_1 = I_c + jI_d$ A if $\bar{I}_s = 10\angle 0^\circ$ A.		b) Bepaal $\bar{I}_1 = I_c + jI_d$ A indien $\bar{I}_s = 10\angle 0^\circ$ A.	
22	V_c	22	V_c
23	V_d	23	V_d
24	I_c	24	I_c
25	I_d	25	I_d

The circuit diagram shows a voltage source V_s in series with a 40Ω resistor. This combination is in parallel with a $-j30\Omega$ capacitor. The voltage across the capacitor is labeled V_{ab} with the positive terminal at the top. The current through the capacitor is labeled I_s with an upward arrow. The current through the 40Ω resistor is labeled I_1 with a downward arrow. The circuit is grounded at the bottom.

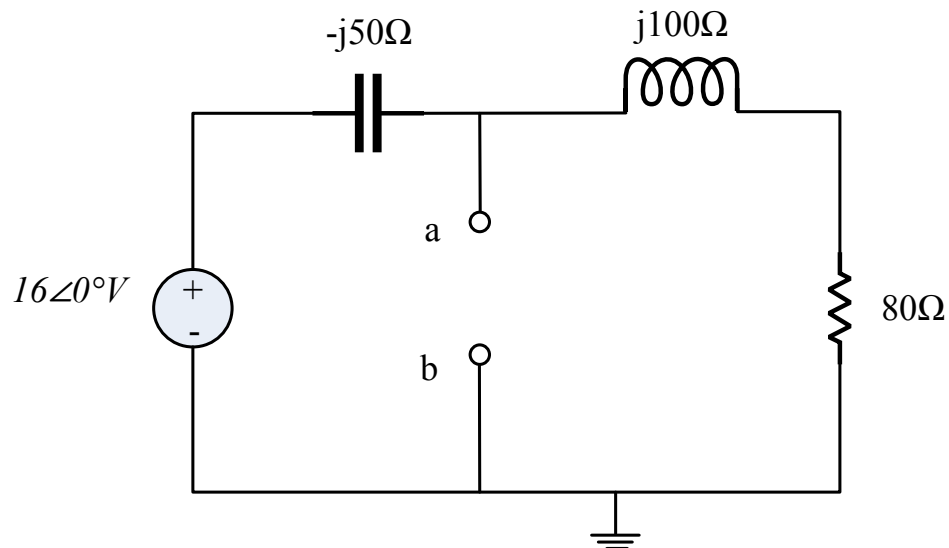
SECTION / AFDELING 4 [16]

Question 26-27 [2]		Vraag 26-27 [2]	
Determine $\bar{V}_0$ in polar form ($C\angle\psi$) V.		Bepaal $\bar{V}_0$ in polêre vorm ($C\angle\psi$) V.	
26	C	26	C
27	ψ	27	ψ

The circuit diagram shows a voltage source $16\angle 0^\circ \text{ V}$ in series with a 10Ω resistor. This is followed by a parallel combination of an inductor $j4\Omega$ and a branch containing a $-j5\Omega$ capacitor in series with a $2\angle -90^\circ \text{ A}$ current source pointing upwards. The voltage $\bar{V}_0$ is measured across the inductor $j4\Omega$. The bottom wire is grounded.

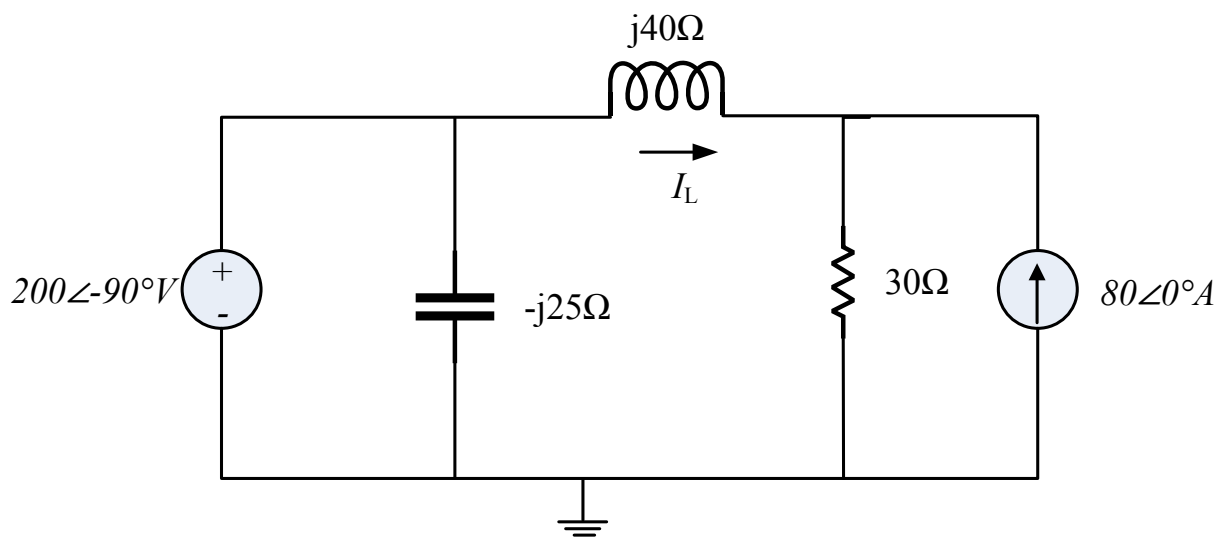
Question 28-33 [4]		Vraag 28-33 [4]	
Use mesh analysis to set up two equations of the following form: $\begin{cases} \bar{I}_1 - \bar{I}_2 = A\angle\theta^\circ \\ a\bar{I}_1 + b\bar{I}_2 = 4\angle 0^\circ \end{cases}$ Give $a = x_1 + jy_1$ and $b = x_2 + jy_2$ in rectangular form , and $A > 0$, $-180^\circ \leq \theta \leq 180^\circ$.		Gebruik lus analyse en formuleer twee vergelykings in die volgende vorm: $\begin{cases} \bar{I}_1 - \bar{I}_2 = A\angle\theta^\circ \\ a\bar{I}_1 + b\bar{I}_2 = 4\angle 0^\circ \end{cases}$ Gee $a = x_1 + jy_1$ en $b = x_2 + jy_2$ in reghoekige vorm , en $A > 0$, $-180^\circ \leq \theta \leq 180^\circ$.	
28	A	28	A
29	θ	29	θ
30	x_1	30	x_1
31	y_1	31	y_1
32	x_2	32	x_2
33	y_2	33	y_2

Question 34-37 [4]		Vraag 34-37 [4]	
Find the Norton equivalent with respect to terminals a - b . Write the answers in rectangular form , where $\bar{I}_N = I_x + jI_y$ A and $Z_N = Z_x + jZ_y \Omega$.		Bepaal die Norton ekwivalent met verwysing tot terminale a - b . Skryf die antwoorde in reghoekige vorm , met $\bar{I}_N = I_x + jI_y$ A en $Z_N = Z_x + jZ_y \Omega$.	
34	I_x	34	I_x
35	I_y	35	I_y
36	Z_x	36	Z_x
37	Z_y	37	Z_y



Question 38-39 [2]		Vraag 38-39 [2]	
Use source transformation to determine $\bar{I}_Y$ in rectangular form $(I_a + jI_b)$ A.		Gebruik brontransformatie om $\bar{I}_Y$ te bepaal in reghoekige vorm $(I_a + jI_b)$ A.	
38	I_a	38	I_a
39	I_b	39	I_b

Question 40-43 [4]		Vraag 40-43 [4]	
a) Determine $\bar{I}_L = I_{(A)a} + jI_{(A)b}$ A only due to the current source.		a) Bepaal $I_{(A)a} + jI_{(A)b}$ A slegs as gevolg van die stroombron.	
b) Determine $\bar{I}_L = I_{(V)a} + jI_{(V)b}$ A only due to the voltage source.		b) Bepaal $\bar{I}_L = I_{(V)a} + jI_{(V)b}$ A slegs as gevolg van die spanningsbron.	
40	$I_{(A)a}$	40	$I_{(A)a}$
41	$I_{(A)b}$	41	$I_{(A)b}$
42	$I_{(V)a}$	42	$I_{(V)a}$
43	$I_{(V)b}$	43	$I_{(V)b}$



PRACTICE PAGE / OEFENBLAD

Student number

--	--	--	--	--	--	--	--

Assessment ID

2	0	1	7	E	B	N	1	1	1	S	3		
---	---	---	---	---	---	---	---	---	---	---	---	--	--

Answer

Unit

01	$V_O =$	V
02	$V_S =$	V
03	$E_L =$	mJ
04	$E_C =$	mJ
05	$a =$	
06	$b =$	
07	$C_{eq} =$	μF
08	$R =$	$m\Omega$
09	$C =$	mF
10	$A =$	A
11	$\phi =$	$^\circ$
12	$Y_a =$	S
13	$Y_b =$	S
14	$a =$	
15	$b =$	
16	$A =$	
17	$\theta =$	$^\circ$
18	$B =$	
19	$\phi =$	$^\circ$
20		
21	$\phi_d =$	$^\circ$
22	$V_c =$	V
23	$V_d =$	V
24	$I_c =$	A
25	$I_d =$	A

26	$C =$	V
27	$\psi =$	$\circ$
28	$A =$	
29	$\theta =$	$\circ$
30	$x_1 =$	
31	$y_1 =$	
32	$x_2 =$	
33	$y_2 =$	
34	$I_x =$	A
35	$I_y =$	A
36	$Z_x =$	Ω
37	$Z_y =$	Ω
38	$I_a =$	A
39	$I_b =$	A
40	$I_{(A)a} =$	A
41	$I_{(A)b} =$	A
42	$I_{(V)a} =$	A
43	$I_{(V)b} =$	A